



WeAreDevelopers · Berlin · 2026

Stop Guessing, Start Measuring.

Evaluating RAG systems with synthetic test data.

SPEAKER

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REALITY CHECK

The reality of building LLM-based chatbots grounded in your data.

DAY 1

Looks great.

You build a chatbot grounded in your data, try it out, every answer looks good — ready for deployment!

TWO WEEKS LATER

Confidently wrong.

It answers a real user question with incorrect facts, but nobody notices it.

There is a low-effort, repeatable way to measure the performance of your system, and catch issues before users do.



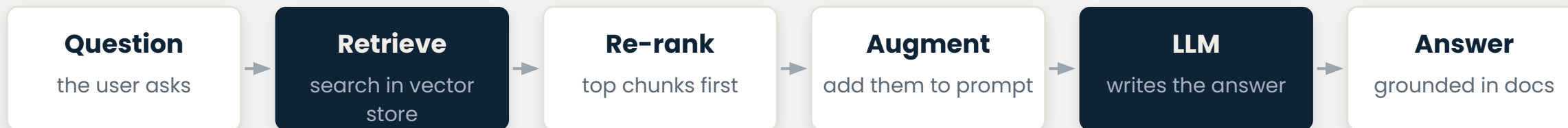
RETRIEVAL AUGMENTED GENERATION

Where RAG can **break**.

PREPARE YOUR KNOWLEDGE • ONCE, OFFLINE



ANSWER A QUERY



RETRIEVAL

Did we fetch the right context?

measureable

GENERATION

Did the LLM use the context well?

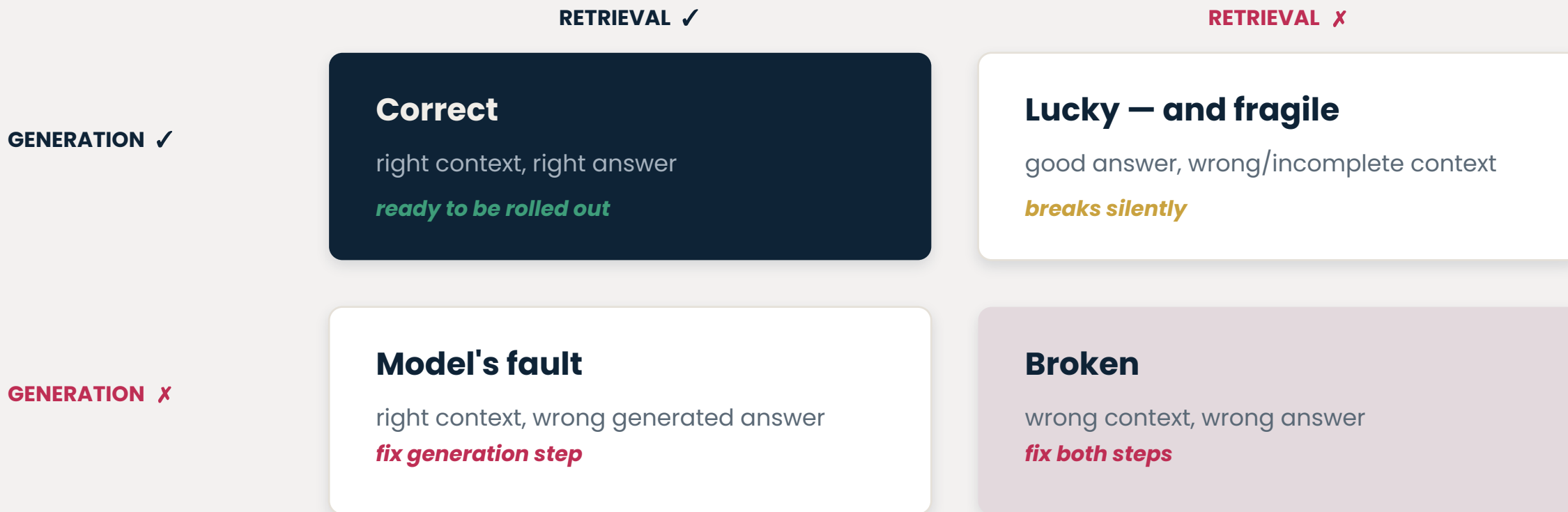
measureable



WHY EVALUATION IS HARD

Two **failure surfaces** for RAG applications.

A good answer can come from bad retrieval — and vice-versa. These should be measured separately.





THE EVAL BLOCKER

Good testing means **volume & coverage**.

- **5 questions $\neq$ coverage**

the cases that break the system are the ones we didn't consider testing

- **Regressions stay invisible**

a prompt update might fix one thing and quietly break another

- **No baseline, no A/B**

can't tell if tweaking a system is helpful if there's no scores serving as comparison

To score a system, we need a labelled set:

question	reference answer	reference context
How to download a HuggingFace model?	In order to download a model, you need to ...	chunk X, chunk Y, chunk Z
?	?	?
?	?	?
?	?	?

Hand-crafting the test set can be slow, expensive, and biased toward the test sample.

Is there a way to handle the volume & coverage requirement?



SYNTHETIC TEST DATA

Ways to generate synthetic data.

How to turn your documents into a labelled test set:



THE NAIVE SHORTCUT

Loop an LLM over each chunk and generate questions.

- ✗ One chunk per question — no complexity
- ✗ Shallow, repetitive questions
- ✗ Answers leak from their own chunk
- ✗ Lack of variety

Volume, not coverage.



RAGAS

Build a knowledge graph from your docs, then generate questions from it.

- ✓ Simple + complex questions
- ✓ Specific and abstract phrasing
- ✓ Reference answer + context for each item
- ✓ Persona-driven variety

Volume and coverage.



TOOLKIT

RAGAS Framework

Open-source evaluation framework for LLM, RAG and agentic applications. Integrates with LangChain, LlamaIndex, and more.



Datasets

Generated or curated test sets.



Experiments

Running a pipeline over a dataset
using an endpoint.



Metrics

Score each experiment
(LLM-as-judge)

Today we use: **(1) synthetic test-set generation** (knowledge-graph based)
(2) the RAG metrics (LLM-as-judge).



COVERAGE

Build a **graph** to get diverse questions.

SPECIFIC

SINGLE-HOP

Specific · single-hop

"What architecture is the tokenizers-linux-x64-musl binary for?"

MULTI-HOP

Specific · multi-hop

"Which retriever does the example use, and what library provides it?"

ABSTRACT

Abstract · single-hop

"What is a tokenizer, in one sentence?"

Abstract · multi-hop

"How do datasets and tokenizers fit together on HuggingFace?"



Why a knowledge graph?

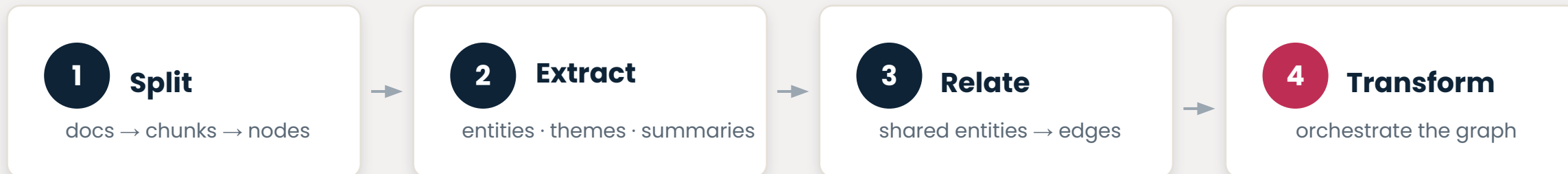
To build a complex question, you first have to know which **chunks** are **related**. A knowledge graph captures exactly those links — so RAGAS can **traverse** it and assemble the **context** each question type needs.



KNOWLEDGE GRAPH

How RAGAS builds a **test set**.

BUILD THE GRAPH



GENERATE QUESTIONS



Steer the mix with personas + a weighted distribution (e.g. 40% entity · 40% theme · 20% multi-hop) so the set resembles your users' behavior.



THE METRICS

Score the **trajectory**, not just the answer.

Every evaluated row has 4 components

question

retrieved_contexts

response

reference

gold label



Retrieval metrics

read contexts vs reference

Context Precision · Context Recall



Generation metrics

read response vs context / reference

Faithfulness · Response Relevancy · Noise Sensitivity

Different metrics assess different parts of the RAG mechanism — we can localize a failure to retrieval vs generation.



RETRIEVAL METRICS

Did it fetch the **right** context?

ranking



Context Precision

Are the relevant chunks ranked at the top?

Rank-aware signal-to-noise.

Low = the retriever returns junk or buries the good.

higher is better

coverage



Context Recall

Did we pull in everything needed?

Coverage — measured against the reference.

Low = the answer needed 3 chunks, you fetched 1.

higher is better

Precision vs recall = the classic trade-off.



GENERATION METRICS

Did it **use** the context well?



Faithfulness

Is every claim grounded in the context?

Anti-hallucination. Catches the confidently-wrong answers.

higher is better



Response Relevancy

Does the answer address the question?

Catches partial or padded answers.

higher is better



Noise Sensitivity

Does junk context throw the LLM off?

Feed relevant or irrelevant chunks — how often does it make a wrong claim?

lower is better



READING SCORES

Read metrics like a **diagnostician**.

	Ctx Prec	Ctx Rec	Faith	Resp Rel	Noise ↓
row 1	0.91	0.42	0.95	0.93	0.08
row 2	0.88	0.86	0.90	0.91	0.10
mean	0.86	0.61	0.92	0.90	0.12

Same data, five lenses — the scores tell you what to change.

● **High faithfulness + relevancy, LOW recall**
the model's fine — retrieval is the bottleneck. Fix chunking / top-k, not the prompt.

● **Low faithfulness**
generation is hallucinating → fix the prompt or model.

● **Noise sensitivity creeping up**
junk chunks are derailing answers → tighten retrieval / add reranking.

Tweak one thing, re-measure, repeat — one change per experiment, so every gain is traceable to what you did.



BEYOND SYNTHETIC DATA

A bootstrap, not a **verdict**.

Synthetic data gets you a working loop on day one. It is not the final step.

1 **LLM-as-judge $\neq$ ground truth**

it has biases, and it can share blind spots with your system if it's the same model family

2 **Generated questions can be shallow**

or leak the answer straight from their own chunk

3 **Reference answers can be wrong**

the gold answer is LLM-generated too

4 **Metrics are proxies**

for what you actually care about: user satisfaction



KEEP IT HONEST

Human stays in the **loop**.

KEEP IT REAL

- ✓ Use a small set of human labels
- ✓ Use a different, strong model as judge
- ✓ Review and prune the generated set
- ✓ Mix in real production queries over time
- ✓ Version your test set

MATURATION TIMELINE



Day 1 • Synthetic test set

cold-start — a simple eval loop with zero production data



Over time • Human-in-the-loop + real traffic

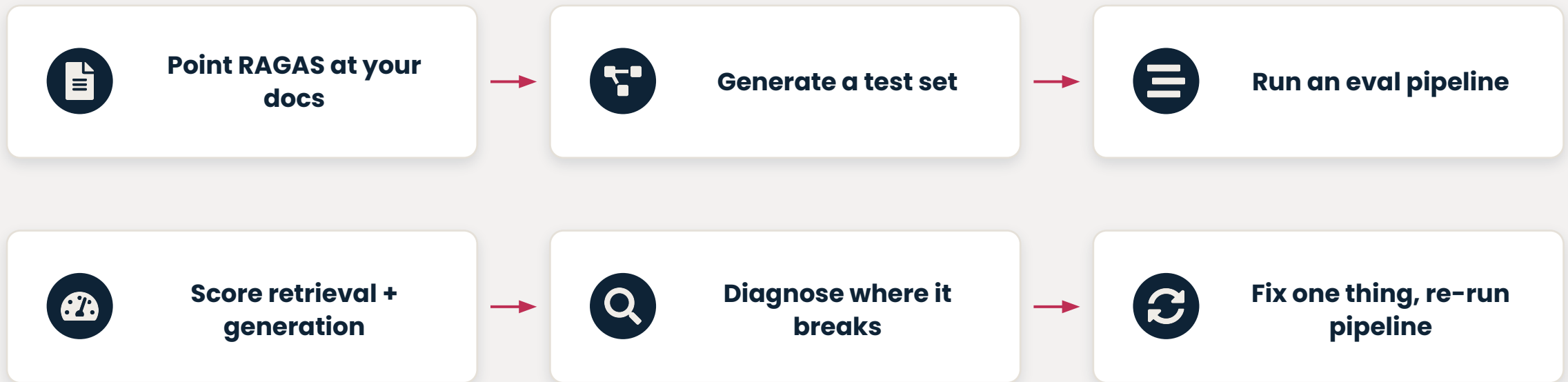
annotate a sample, fold in production queries — the set grows

Using synthetic data is the first step on the roadmap of RAG evaluation.



THE WORKFLOW

Recap.



Layer human review + real queries over time. **Think of evals early, use synthetic data if needed.**



Thank you.

If you have any questions, feel free to reach out to me.

RESOURCES

- **RAGAS docs** docs.ragas.io
- **Demo corpus** huggingface.co/datasets/m-ric/huggingface_doc
- **Notebook + slides** github.com/csenge-szabo/Synthetic_Data_Generation_w_RAGAS

